

PrometheusMySQLMySQL

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rancherPrometheusgrafanaMySQL MHA

mysql exporterMySQL

1.ServiceMonitor

k8smasterServiceMonitorcattle-monitoring-system

mysql-exporter-ServiceMonitor.yaml

```
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  labels:
    app: prometheus-mysql-exporter
    name: prometheus-mysql-exporter
    namespace: cattle-monitoring-system
spec:
  endpoints:
    - interval: 30s
      relabelings:
        - regex: (.*)-exporter
          sourceLabels:
            - __meta_kubernetes_pod_container_name
          targetLabel: instance
      targetPort: 9104
  selector:
    matchLabels:
      app: prometheus-mysql-exporter

# ServiceMonitor
[root@kube-master01 ~]# kubectl -ncattle-monitoring-system create -f mysql-exporter-ServiceMonitor.yaml
servicemonitor.monitoring.coreos.com/prometheus-mysql-exporter created
[root@kube-master01 ~]# kubectl -ncattle-monitoring-system get ServiceMonitor
NAME                                     AGE
prometheus-mysql-exporter              13s
rancher-monitoring-alertmanager        14d
rancher-monitoring-grafana             14d
rancher-monitoring-kube-state-metrics  14d
rancher-monitoring-operator            14d
rancher-monitoring-prometheus          14d
rancher-monitoring-prometheus-node-exporter  14d
```

2.service

mysql-exporter-svc.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: prometheus-mysql-exporter
  namespace: cattle-monitoring-system
  labels:
    app: prometheus-mysql-exporter
  annotations:
    prometheus.io/scrape: "true"
spec:
  ports:
    - name: mysql-exporter
      port: 9104
      protocol: TCP
  selector:
    app: prometheus-mysql-exporter
```

3. Deployment MySQL

mysql-exporter-deploy.yaml

```
#
CREATE user 'prom'@'%' IDENTIFIED by 'K9SFmY@FDprom2024';
GRANT SELECT, PROCESS, REPLICATION CLIENT ON *.* TO 'prom'@'%';

# exporter IP
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: prometheus-mysql-exporter
    name: mysql-his-mdsn01-exporter
    namespace: cattle-monitoring-system
spec:
  replicas: 1
  selector:
    matchLabels:
      app: prometheus-mysql-exporter
  template:
    metadata:
      labels:
        app: prometheus-mysql-exporter
        name: mysql-his-mdsn01-exporter
        namespace: cattle-monitoring-system
    spec:
      containers:
        - args:
            - --collect.info_schema.innodb_metrics
            - --collect.info_schema.tables
            - --collect.info_schema.processlist
            - --collect.info_schema.userstats
            - --collect.info_schema.tables.databases=*
            - --collect.engine_innodb_status
            - --collect.perf_schema.eventsstatements
            - --collect.auto_increment.columns
            - --collect.binlog_size
          command:
            - /bin/mysqld_exporter
          env:
            - name: DATA_SOURCE_NAME
              value: prom:K9SFmY@FDprom2024@(10.20.41.64:3306)/
          image: harbor.rubikstack.com/wowjoy/library/mysqld-exporter:latest
          imagePullPolicy: IfNotPresent
          name: mysql-his-mdsn01-exporter
          ports:
            - containerPort: 9104
              name: 9104tcp
              protocol: TCP
          dnsPolicy: ClusterFirst
          restartPolicy: Always
```

4.PrometheusRule

mysql-alerting.yaml

```
apiVersion: monitoring.coreos.com/v1
kind: PrometheusRule
metadata:
  labels:
    app: rancher-monitoring
    release: rancher-monitoring
```

```

name: prometheus-mysql-alerting
namespace: cattle-monitoring-system
spec:
  groups:
    - name: mysql-alerting
      rules:
        - alert: MysqlDown
          annotations:
            description: Mysql {{ $labels.instance }} Down
            expr: mysql_up == 0
            for: 5m
            labels:
              severity: critical
        - alert: TooManyAbortedConnections
          annotations:
            description: MySQL {{ $labels.instance }} generated {{ printf "%.2f" $value }} Aborted_Connections
            within 5 minute.
            expr: increase(mysql_global_status_aborted_connects[5m]) > 10
            for: 5m
            labels:
              severity: warning
        - alert: TooManySlowQueriesFromMysql
          annotations:
            description: Mysql {{ $labels.instance }} have too many slow queries.
            expr: rate(mysql_global_status_slow_queries[5m]) > 1
            for: 5m
            labels:
              severity: warning
        - alert: TooManySlowQueriesIn1Min
          annotations:
            description: MySQL {{ $labels.instance }} {{ printf "%.2f" $value }}.
            expr: increase(mysql_global_status_slow_queries[1m]) > 50
            for: 2m
            labels:
              severity: warning
        - alert: MysqlOpenFilesHigh
          annotations:
            description: Mysql {{ $labels.instance }} OpenFiles Used is above 70% Current usage is {{ printf "%.2f"
            $value }}%.
            expr: mysql_global_status_innodb_num_open_files / mysql_global_variables_open_files_limit * 100 > 70
            for: 5m
            labels:
              severity: warning
        - alert: MysqlUseTooManyConnections
          annotations:
            description: Mysql {{ $labels.instance }} Connections Used is above 80% Current usage is {{ printf "%.
            2f" $value }}%.
            expr: mysql_global_status_threads_connected / mysql_global_variables_max_connections * 100 > 80
            for: 5m
            labels:
              severity: warning
        - alert: MysqlHaveTooManyAliveConnections
          annotations:
            description: Mysql {{ $labels.instance }} have too many alive connections Current value:{{ $value }}.
            expr: mysql_global_status_threads_running > 50
            for: 3m
            labels:
              severity: critical
        - alert: MysqlHighQPS
          annotations:
            description: Mysql {{ $labels.instance }} QPS high Current QPS:{{ printf "%.2f" $value }}.
            expr: rate(mysql_global_status_questions[5m]) > 8000
            for: 5m
            labels:
              severity: warning
        - alert: SlaveIOThreadStopped
          annotations:
            description: Mysql {{ $labels.instance }} slave io thread stopped.
            expr: mysql_slave_status_slave_io_running != 1
            for: 1m
            labels:

```

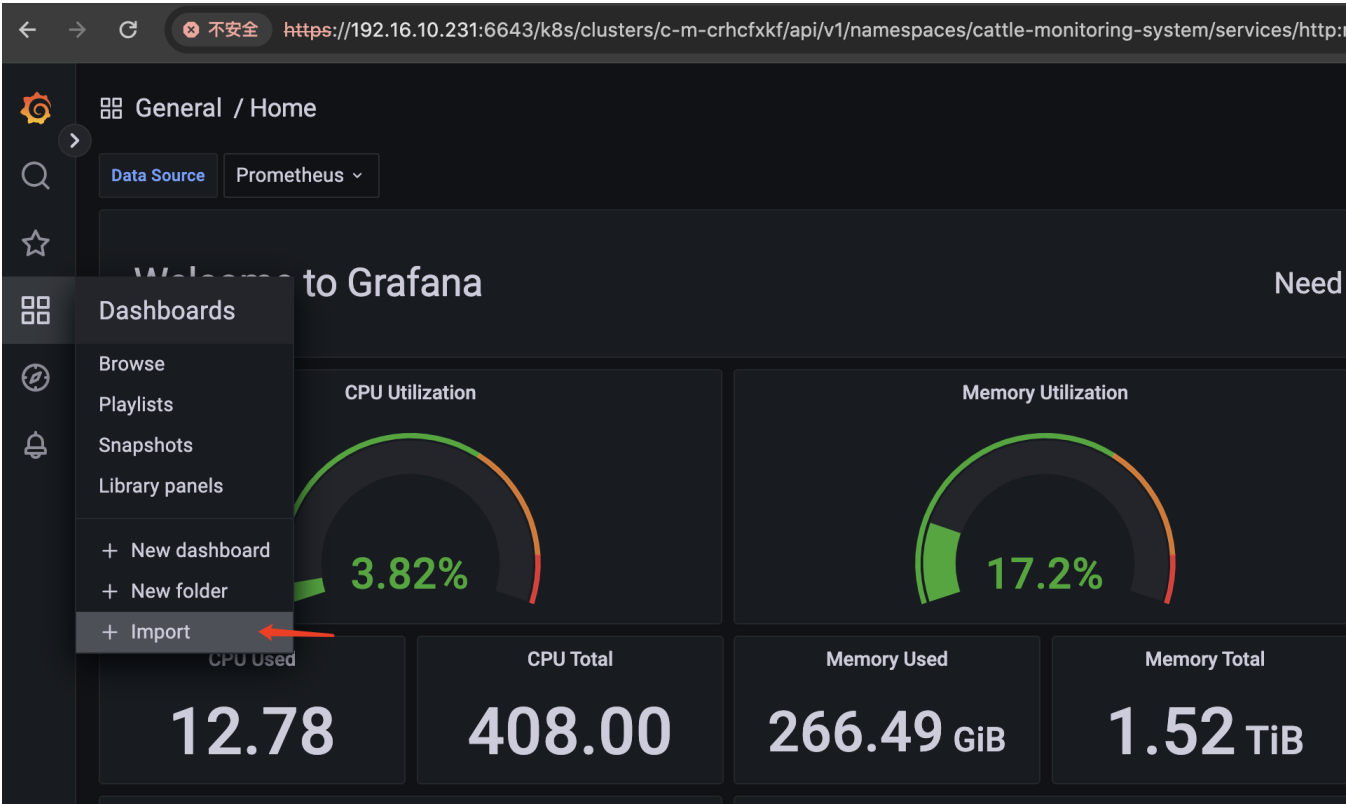
```

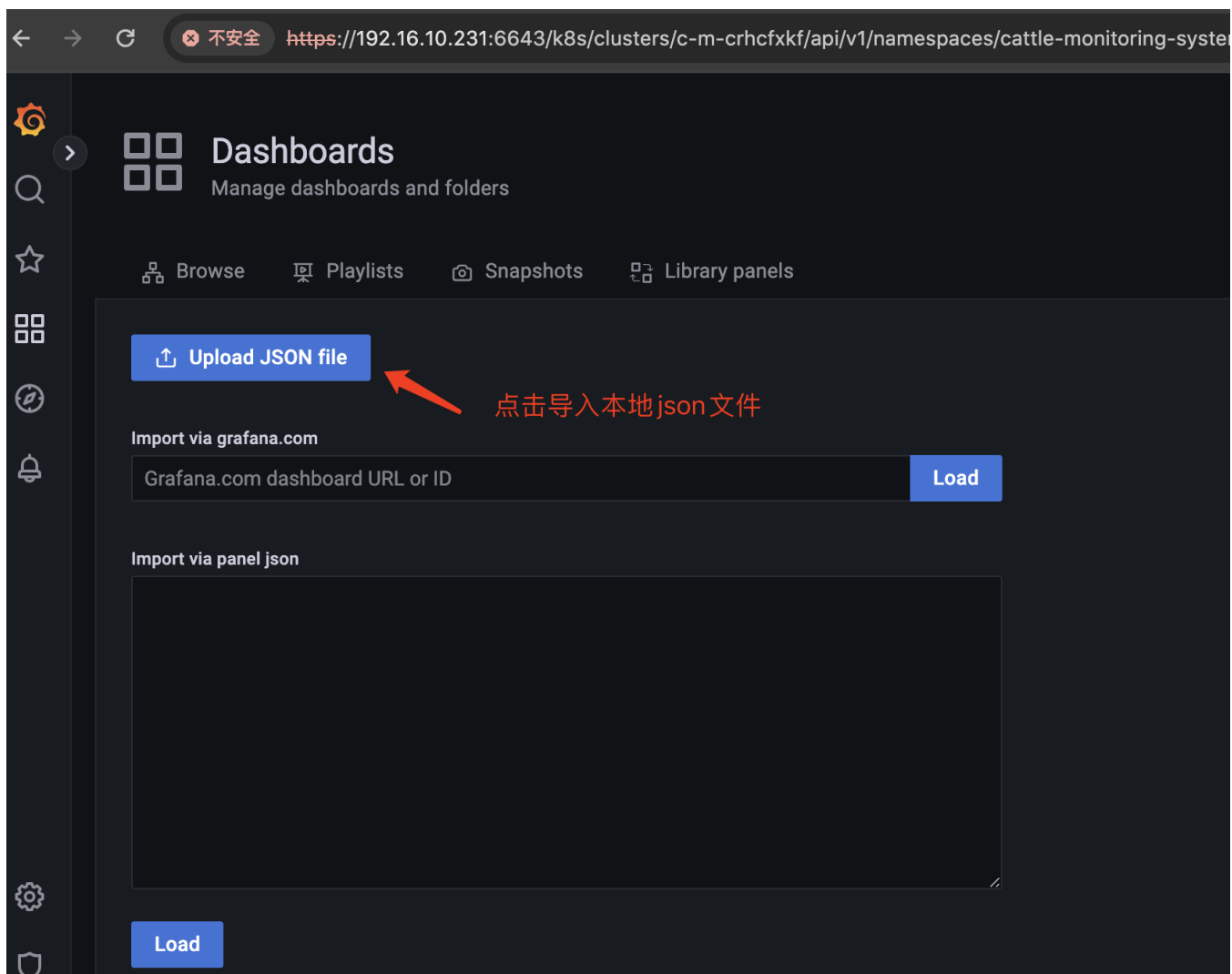
    severity: critical
- alert: SlaveSQLThreadStopped
  annotations:
    description: Mysql {{ $labels.instance }} sql io thread stopped.
    expr: mysql_slave_status_slave_sql_running != 1
    for: 1m
    labels:
      severity: critical
- alert: SlaveLaggingBehindMaster60s
  annotations:
    description: Mysql {{ $labels.instance }} seconds_behind_master > 60s Current value:{{ $value }}s.
    expr: mysql_slave_status_seconds_behind_master > 60
    for: 2m
    labels:
      severity: warning
- alert: MySQLAutoIncrementUsageHigh
  expr: (mysql_info_schema_auto_increment_column / mysql_info_schema_auto_increment_column_max) * 100 > 80
  for: 5m
  labels:
    severity: warning
  annotations:
    description: The auto increment usage on {{ $labels.schema }}.{{ $labels.table }} is above 80%. Current
usage is {{ printf "%.2f" $value }}%.
- alert: Mysql_Deadlock
  expr: increase(mysql_info_schema_innodb_metrics_lock_lock_deadlocks_total[1m]) > 0
  labels:
    severity: warning
  annotations:
    description: Mysql {{ $labels.instance }} Deadlock was found NumberIn1Min:{{ printf "%.2f" $value }}.
- alert: Mysql_Instance_Reboot
  expr: mysql_global_status_uptime < 180
  for: 1m
  labels:
    severity: critical
  annotations:
    description: Mysql {{ $labels.instance }} Instance_Reboot in 3 minute Current uptime:{{ $value }}
seconds.

```

5.grafana

k8s masterkubectlranchergrafanaMySQL







mysql-overview_rev5_wowjoy.json



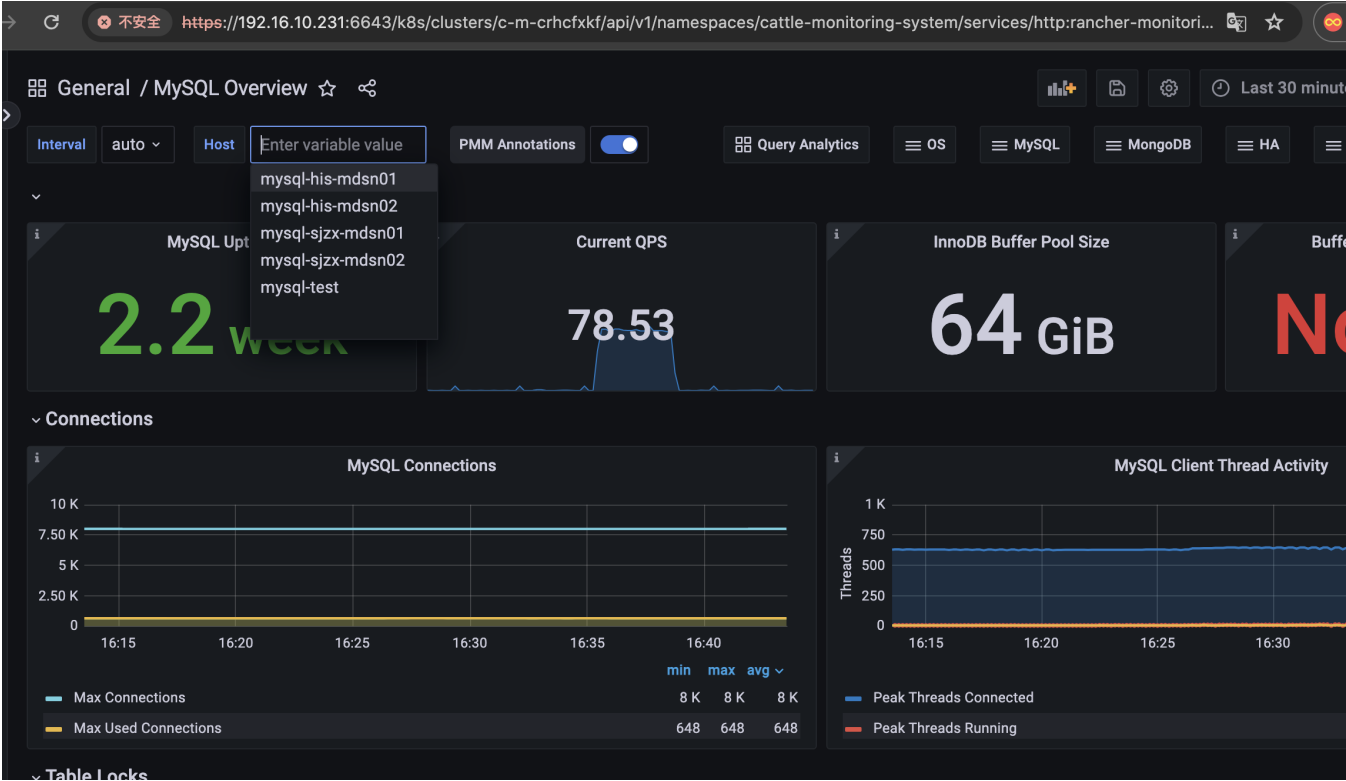
mysql-replication_rev1_wowjoy.json



Mysqld_Exporter_...rev3_wowjoy.json



mysql-innodb-met...rev1_wowjoy.json



6.

Prometheus

Prometheus

Alerts

Graph

Status

Help

Targets

AllUnhealthyExpand All

serviceMonitor/cattle-monitoring-system/prometheus-mysql-exporter/0 (5/5 up) show less

Endpoint	UP	Labels	Last Scrape	Scrape Duration
http://10.43.33.42:9104/metrics	UP	container="mysql-test-exporter" endpoint="9104" instance="mysql-test" job="prometheus-mysql-exporter" namespace="cattle-monitoring-system" pod="mysql-test-exporter-b67bdcc46-mkdhg" service="prometheus-mysql-exporter"	42.590s ago	34.057ms
http://10.43.134.44:9104/metrics	UP	container="mysql-his-mdsn01-exporter" endpoint="9104" instance="mysql-his-mdsn01" job="prometheus-mysql-exporter" namespace="cattle-monitoring-system" pod="mysql-his-mdsn01-exporter-788d6875d8-lrhkr" service="prometheus-mysql-exporter"	37.990s ago	29.277ms
http://10.43.134.12:9104/metrics	UP	container="mysql-his-mdsn02-exporter" endpoint="9104" instance="mysql-his-mdsn02" job="prometheus-mysql-exporter" namespace="cattle-monitoring-system" pod="mysql-his-mdsn02-exporter-6d96fd6957-hn5xr"	24.521s ago	30.233ms

Runtime & Build Information

TSDB Status

Command-Line Flags

Configuration

Rules

Targets

Service Discovery

进入 target 页面 查看 mysql-exporter 是否启动

← → ↻ 不安全 https://192.16.10.231:6643/k8s/clusters/c-m-crhcfxkf/api/v1/namespaces/cattle-monitoring-system/services/http.rancher-monitori... ☆

Prometheus Alerts Graph Status ▾ Help

Rules

mysql-alerting

49.726s a

Rule	State	Error	Last Evaluation
alert: MysqlDown expr: mysql_up == 0 for: 5m labels: severity: critical annotations: message: Mysql {{ \$labels.instance }} Down	OK		49.727s ago
alert: TooManyErrorFromMysql expr: sum(irate(mysql_global_status_connection_errors_total[1m])) > 10 for: 5m labels: severity: warning annotations: message: Mysql {{ \$labels.instance }} have too many error.	OK		49.726s ago
alert: TooManySlowQueriesFromMysql expr: rate(mysql_global_status_slow_queries[5m]) > 1 for: 5m labels:	OK		49.726s ago

Runtime & Build Information

TSDB Status

Command-Line Flags

Configuration

Rules

Targets

Service Discovery

进入 rules 页面 确认规则生效

node-exporter

1.node-exporternode-exporter

node-exporter/root/



node_exporter-1....nux-amd64.tar.gz

```
#
tar -xzf node_exporter-1.5.0.linux-amd64.tar.gz
chown -R root:root node_exporter-1.5.0.linux-amd64

#node_exporter
[root@node ~]# vi /usr/lib/systemd/system/node_exporter.service
[Unit]
Description=prometheus node_exporter

[Service]
Type=simple
ExecStart=/root/node_exporter-1.5.0.linux-amd64/node_exporter
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=on-failure
RestartSec=60s
MemoryLimit=300M
CPUQuota=100%

[Install]
WantedBy=multi-user.target

#
systemctl daemon-reload
systemctl start node_exporter
systemctl enable node_exporter
```

2.ServiceMonitorServiceEndpointPrometheusRule

k8smastercattle-monitoring-systemmonitoring

mysql-node-exporter-servicemonitor.yaml

```
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  labels:
    app: prometheus-mysql-node-exporter
  name: mysql-node-exporter
  namespace: cattle-monitoring-system
spec:
  jobLabel: mysql-node-exporter
  endpoints:
    - port: mysql-exporter
      interval: 15s
      honorLabels: true
      scheme: http
      relabelings:
        - action: replace
          regex: (.*)
          replacement: $1
          sourceLabels:
            - __meta_kubernetes_endpoint_node_name
          targetLabel: instance
  selector:
    matchLabels:
      app: prometheus-mysql-node-exporter
```

mysql-node-exporter-svc.yaml

```
apiVersion: v1
kind: Service
metadata:
  annotations:
    prometheus.io/scrape: "true"
  labels:
    app: prometheus-mysql-node-exporter
  name: prometheus-mysql-node-exporter
  namespace: cattle-monitoring-system
spec:
  ports:
    - name: mysql-exporter
      port: 9100
      protocol: TCP
      targetPort: 9100
  type: ClusterIP
```

mysql-node-exporter-ep.yaml

```
apiVersion: v1
kind: Endpoints
metadata:
  labels:
    app: prometheus-mysql-node-exporter
    name: prometheus-mysql-node-exporter
    namespace: cattle-monitoring-system
subsets:
- addresses:
  - ip: 192.168.190.41
    nodeName: his-mdsn01
  - ip: 192.168.190.42
    nodeName: his-mdsn02
  - ip: 192.168.190.43
    nodeName: his-mdsn03
  - ip: 192.168.190.44
    nodeName: his-mha
  - ip: 192.168.190.51
    nodeName: sjzx-mdsn01
  - ip: 192.168.190.52
    nodeName: sjzx-mdsn02
  - ip: 192.168.190.53
    nodeName: sjzx-mdsn03
  - ip: 192.168.190.54
    nodeName: sjzx-mha
  - ip: 192.168.190.202
    nodeName: xtrabackup
  - ip: 192.168.190.90
    nodeName: dsmart
  - ip: 192.168.190.81
    nodeName: mongo01
  - ip: 192.168.190.82
    nodeName: mongo02
  - ip: 192.168.190.83
    nodeName: mongo-zc
  - ip: 192.168.190.201
    nodeName: mysql-test
  - ip: 192.168.190.203
    nodeName: mongo-test
  - ip: 192.168.190.101
    nodeName: canal01
  - ip: 192.168.190.102
    nodeName: canal02
  - ip: 192.168.190.103
    nodeName: canal03
  - ip: 192.168.190.111
    nodeName: sjck01
  - ip: 192.168.190.112
    nodeName: sjck02
  - ip: 192.168.190.113
    nodeName: sjck03
  ports:
  - name: mysql-exporter
    port: 9100
    protocol: TCP
```

mysql-node-exporter-prometheusrule.yaml

```
apiVersion: monitoring.coreos.com/v1
kind: PrometheusRule
metadata:
  labels:
    prometheus: k8s
    role: alert-rules
  name: mysql-node-alerting
```

```

namespace: cattle-monitoring-system
spec:
  groups:
    - name: mysql-node-alerting
      rules:
        - alert: NodeDown
          expr: up{job="prometheus-mysql-node-exporter"} == 0
          for: 1m
          labels:
            severity: critical
          annotations:
            description: ": {{ $labels.instance }} ."
        - alert: NodeFilesystemIsLessThan10%
          annotations:
            description: Filesystem on {{ $labels.mountpoint }} at {{ $labels.instance }}
              has only {{ printf "%.2f" $value }}% available space left.
            summary: Filesystem has less than 10% space left.
          expr: node_filesystem_free_bytes{fstype=~"ext4|xfs",job="prometheus-mysql-node-exporter",mountpoint!~".
*tmp|.boot" }/node_filesystem_size_bytes{fstype=~"ext4|xfs",job="prometheus-mysql-node-exporter",mountpoint!~".
*tmp|.boot" } * 100 < 10
          for: 30m
          labels:
            severity: critical
        - alert: NodeMemUsedHigh
          annotations:
            description: Node {{ $labels.instance }} Memory Utilisation has more then 90% Current value:{{ printf
            "%.2f" $value }}%.
          expr: '(1 - (node_memory_MemAvailable_bytes{job="prometheus-mysql-node-exporter"} /
node_memory_MemTotal_bytes{job="prometheus-mysql-node-exporter"}))*100 > 90'
          for: 5m
          labels:
            severity: critical
        - alert: NodeCPUHigh
          annotations:
            description: Node {{ $labels.instance }} CPU Utilisation has more then 80% Current value:{{ printf "%.
            2f" $value }}%.
          expr: ((1 - avg(irate(node_cpu_seconds_total{mode="idle",job="prometheus-mysql-node-exporter"}[1m])) by
(instance)))*100 > 80
          for: 1m
          labels:
            severity: critical
        - alert: NodeProcessHigh
          annotations:
            description: Node {{ $labels.instance }} has too many running processes.
          expr: sum by (process_name,instance) (rate(process_cpu_seconds_total{job="prometheus-mysql-node-
exporter", mode!="idle"}[1m])) * 100 > 90
          for: 10m
          labels:
            severity: critical
        - alert: NodeFileDescriptorsHigh
          annotations:
            description: Node {{ $labels.instance }} has too many file descriptors.
          expr: sum(node_filefd_allocated{job="prometheus-mysql-node-exporter"}) by (instance) / 64000 * 100 > 90
          for: 10m
          labels:
            severity: warning
        - alert: NodeLoad1High
          annotations:
            description: Node {{ $labels.instance }} 1min avg load has more then 80% of the node load capacity.
          expr: (sum(node_load1{job="prometheus-mysql-node-exporter"}) by (instance)) / (count
(node_cpu_seconds_total{job="prometheus-mysql-node-exporter",mode="system"})by (instance)) > 0.8
          for: 1m
          labels:
            severity: warning
        - alert: NodeLoad5High
          annotations:
            description: Node {{ $labels.instance }} 5min avg load has more then 80% of the node load capacity.
          expr: (sum(node_load5{job="prometheus-mysql-node-exporter"}) by (instance)) / (count
(node_cpu_seconds_total{job="prometheus-mysql-node-exporter",mode="system"})by (instance)) > 0.8
          for: 1m
          labels:

```

```
severity: warning
- alert: NodeLoad15High
  annotations:
    description: Node {{ $labels.instance }} 15min avg load has more then 80% of the node load capacity.
    expr: (sum(node_load15{job="prometheus-mysql-node-exporter"}) by (instance)) / (count
(node_cpu_seconds_total{job="prometheus-mysql-node-exporter",mode="system"})by (instance)) > 0.8
    for: 1m
    labels:
      severity: warning
```

yaml kubectl -ncattle-monitoring-system create -f xxx.yaml EndpointsIP

3.

rancherPrometheusgrafana

Prometheus Alerts Graph Status Help

serviceMonitor/cattle-monitoring-system/mysql-node-exporter/0 (14/20 up) show less

Endpoint	State	Labels	Last Scrape	Duration	Error
http://192.168.190.41:9100/metrics	DOWN	endpoint="mysql-exporter" instance="his-mdsn01" job="prometheus-mysql-node-exporter" namespace="cattle-monitoring-system" service="prometheus-mysql-node-exporter"	19.744s ago	10.0s	Get "http://192.168.190.41:9100/metrics": con adline exceeded
http://192.168.190.101:9100/metrics	UP	endpoint="mysql-exporter" instance="canal01" job="prometheus-mysql-node-exporter" namespace="cattle-monitoring-system" service="prometheus-mysql-node-exporter"	8.852s ago	17.115ms	
http://192.168.190.81:9100/metrics	UP	endpoint="mysql-exporter" instance="mongo01" job="prometheus-mysql-node-exporter" namespace="cattle-monitoring-system" service="prometheus-mysql-node-exporter"	3.700s ago	22.331ms	
http://192.168.190.201:9100/metrics	UP	endpoint="mysql-exporter" instance="mysql-test" job="prometheus-mysql-node-exporter" namespace="cattle-monitoring-system" service="prometheus-mysql-node-exporter"	14.645s ago	18.027ms	
http://192.168.190.102:9100/metrics	UP	endpoint="mysql-exporter" instance="canal02" job="prometheus-mysql-node-exporter" namespace="cattle-monitoring-system" service="prometheus-mysql-node-exporter"	9.270s ago	16.821ms	

选择target页面 确认node-exporter是否都启动，若有down掉的 则要排查node-exporter状态及网络问题

Prometheus Alerts Graph Status ▾ Help

annotations:
current_value: {{ .Value }}

选择 rules 确认告警规则是否生效

mysql-node-alerting

7.502s ago22.68

Rule	State	Error	Last Evaluation	Evaluation Time
alert: NodeDown expr: up{job="prometheus-mysql-node-exporter"} == 0 for: 1m labels: severity: critical annotations: description: 主机: [{{ \$labels.instance }}] 服务器宕机.	OK		7.502s ago	1.911ms
alert: NodeFilesystemIsLessThan10% expr: node_filesystem_free_bytes{fstype=~"ext4 xfs",job="prometheus-mysql-node-exporter",mountpoint!~".*tmp .boot"} / node_filesystem_size_bytes{fstype=~"ext4 xfs",job="prometheus-mysql-node-exporter",mountpoint!~".*tmp .boot"} * 100 < 10 for: 30m labels: severity: critical annotations: description: Filesystem on [{{ \$labels.mountpoint }}] at [{{ \$labels.instance }}] has only [{{ printf "%.2f" \$value }}]% available space left.	OK		7.500s ago	10.377ms

grafanadashboard

MySQL-Nodes-Exporter.json

```
{
  "annotations": {
    "list": [
      {
        "builtIn": 1,
        "datasource": {
          "type": "grafana",
          "uid": "-- Grafana --"
        },
        "enable": true,
        "hide": true,
        "iconColor": "rgba(0, 211, 255, 1)",
        "name": "Annotations & Alerts",
        "target": {
          "limit": 100,
          "matchAny": false,
          "tags": [],
          "type": "dashboard"
        },
        "type": "dashboard"
      }
    ]
  },
  "editable": false,
  "fiscalYearStartMonth": 0,
  "graphTooltip": 1,
  "id": 46,
  "links": [],
  "liveNow": false,
  "panels": [
    {
      "collapsed": false,
      "datasource": {
        "type": "datasource",
        "uid": "grafana"
      },
      "gridPos": {
```

```

        "h": 1,
        "w": 24,
        "x": 0,
        "y": 0
    },
    "id": 10,
    "panels": [],
    "targets": [
        {
            "datasource": {
                "type": "datasource",
                "uid": "grafana"
            },
            "refId": "A"
        }
    ],
    "title": "CPU",
    "type": "row"
},
{
    "aliasColors": {},
    "bars": false,
    "dashLength": 10,
    "dashes": false,
    "datasource": {
        "uid": "$datasource"
    },
    "fill": 1,
    "fillGradient": 0,
    "gridPos": {
        "h": 7,
        "w": 12,
        "x": 0,
        "y": 1
    },
    "hiddenSeries": false,
    "id": 2,
    "legend": {
        "alignAsTable": false,
        "avg": false,
        "current": false,
        "max": false,
        "min": false,
        "rightSide": false,
        "show": true,
        "total": false,
        "values": false
    },
    "lines": true,
    "linewidth": 1,
    "links": [],
    "nullPointMode": "null",
    "options": {
        "alertThreshold": true
    },
    "percentage": false,
    "pluginVersion": "9.1.5",
    "pointradius": 5,
    "points": false,
    "renderer": "flot",
    "seriesOverrides": [],
    "spaceLength": 10,
    "stack": true,
    "steppedLine": false,
    "targets": [
        {
            "datasource": {
                "uid": "$datasource"
            },
            "expr": "(\n (1 - sum without (mode) (rate(node_cpu_seconds_total{job=\"prometheus-mysql-node-exporter\", mode=~\"idle|iowait|steal\", instance=\"$instance\"}[$__rate_interval])))\n/ ignoring(cpu)

```

```

group_left\n count without (cpu, mode) (node_cpu_seconds_total{job=\"prometheus-mysql-node-exporter\", mode=\"
idle\", instance=\"$instance\"})\n)\n\n",
    "format": "time_series",
    "intervalFactor": 5,
    "legendFormat": "{{cpu}}",
    "refId": "A"
  }
],
"thresholds": [],
"timeRegions": [],
"title": "CPU Usage",
"tooltip": {
  "shared": true,
  "sort": 0,
  "value_type": "individual"
},
"type": "graph",
"xaxis": {
  "mode": "time",
  "show": true,
  "values": []
},
"yaxes": [
  {
    "format": "percentunit",
    "logBase": 1,
    "max": 1,
    "min": 0,
    "show": true
  },
  {
    "format": "percentunit",
    "logBase": 1,
    "max": 1,
    "min": 0,
    "show": true
  }
],
"yaxis": {
  "align": false
}
},
{
  "aliasColors": {},
  "bars": false,
  "dashLength": 10,
  "dashes": false,
  "datasource": {
    "uid": "$datasource"
  },
  "fill": 0,
  "fillGradient": 0,
  "gridPos": {
    "h": 7,
    "w": 12,
    "x": 12,
    "y": 1
  },
  "hiddenSeries": false,
  "id": 3,
  "legend": {
    "alignAsTable": false,
    "avg": false,
    "current": false,
    "max": false,
    "min": false,
    "rightSide": false,
    "show": true,
    "total": false,
    "values": false
  },

```

```

"lines": true,
"linewidth": 1,
"links": [],
"nullPointMode": "null",
"options": {
  "alertThreshold": true
},
"percentage": false,
"pluginVersion": "9.1.5",
"pointradius": 5,
"points": false,
"renderer": "flot",
"seriesOverrides": [],
"spaceLength": 10,
"stack": false,
"steppedLine": false,
"targets": [
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "node_load1{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\"}",
    "format": "time_series",
    "intervalFactor": 2,
    "legendFormat": "1m load average",
    "refId": "A"
  },
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "node_load5{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\"}",
    "format": "time_series",
    "intervalFactor": 2,
    "legendFormat": "5m load average",
    "refId": "B"
  },
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "node_load15{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\"}",
    "format": "time_series",
    "intervalFactor": 2,
    "legendFormat": "15m load average",
    "refId": "C"
  },
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "count(node_cpu_seconds_total{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\", mode=\"idle\"})",
    "format": "time_series",
    "intervalFactor": 2,
    "legendFormat": "logical cores",
    "refId": "D"
  }
],
"thresholds": [],
"timeRegions": [],
"title": "Load Average",
"tooltip": {
  "shared": true,
  "sort": 0,
  "value_type": "individual"
},
"type": "graph",
"xaxis": {
  "mode": "time",
  "show": true,

```

```

    "values": []
  },
  "yaxes": [
    {
      "format": "short",
      "logBase": 1,
      "min": 0,
      "show": true
    },
    {
      "format": "short",
      "logBase": 1,
      "min": 0,
      "show": true
    }
  ],
  "yaxis": {
    "align": false
  }
},
{
  "collapsed": false,
  "datasource": {
    "type": "datasource",
    "uid": "grafana"
  },
  "gridPos": {
    "h": 1,
    "w": 24,
    "x": 0,
    "y": 8
  },
  "id": 11,
  "panels": [],
  "targets": [
    {
      "datasource": {
        "type": "datasource",
        "uid": "grafana"
      },
      "refId": "A"
    }
  ],
  "title": "Memory",
  "type": "row"
},
{
  "aliasColors": {},
  "bars": false,
  "dashLength": 10,
  "dashes": false,
  "datasource": {
    "uid": "$datasource"
  },
  "fill": 1,
  "fillGradient": 0,
  "gridPos": {
    "h": 7,
    "w": 18,
    "x": 0,
    "y": 9
  },
  "hiddenSeries": false,
  "id": 4,
  "legend": {
    "alignAsTable": false,
    "avg": false,
    "current": false,
    "max": false,
    "min": false,
    "rightSide": false,

```

```

    "show": true,
    "total": false,
    "values": false
  },
  "lines": true,
  "linewidth": 1,
  "links": [],
  "nullPointMode": "null",
  "options": {
    "alertThreshold": true
  },
  "percentage": false,
  "pluginVersion": "9.1.5",
  "pointradius": 5,
  "points": false,
  "renderer": "flot",
  "seriesOverrides": [],
  "spaceLength": 10,
  "stack": true,
  "steppedLine": false,
  "targets": [
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "(\n  node_memory_MemTotal_bytes{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\"}\n-\n  node_memory_MemFree_bytes{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\"}\n-\n  node_memory_Buffers_bytes{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\"}\n-\n  node_memory_Cached_bytes{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\"}\n)\n",
      "format": "time_series",
      "intervalFactor": 2,
      "legendFormat": "memory used",
      "refId": "A"
    },
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "node_memory_Buffers_bytes{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\"}",
      "format": "time_series",
      "intervalFactor": 2,
      "legendFormat": "memory buffers",
      "refId": "B"
    },
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "node_memory_Cached_bytes{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\"}",
      "format": "time_series",
      "intervalFactor": 2,
      "legendFormat": "memory cached",
      "refId": "C"
    },
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "node_memory_MemFree_bytes{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\"}",
      "format": "time_series",
      "intervalFactor": 2,
      "legendFormat": "memory free",
      "refId": "D"
    }
  ],
  "thresholds": [],
  "timeRegions": [],
  "title": "Memory Usage",
  "tooltip": {
    "shared": true,

```

```

    "sort": 0,
    "value_type": "individual"
  },
  "type": "graph",
  "xaxis": {
    "mode": "time",
    "show": true,
    "values": []
  },
  "yaxes": [
    {
      "format": "bytes",
      "logBase": 1,
      "min": 0,
      "show": true
    },
    {
      "format": "bytes",
      "logBase": 1,
      "min": 0,
      "show": true
    }
  ],
  "yaxis": {
    "align": false
  }
},
{
  "datasource": {
    "uid": "$datasource"
  },
  "fieldConfig": {
    "defaults": {
      "mappings": [],
      "max": 100,
      "min": 0,
      "thresholds": {
        "mode": "absolute",
        "steps": [
          {
            "color": "rgba(50, 172, 45, 0.97)",
            "value": null
          },
          {
            "color": "rgba(237, 129, 40, 0.89)",
            "value": 80
          },
          {
            "color": "rgba(245, 54, 54, 0.9)",
            "value": 90
          }
        ]
      }
    },
    "unit": "percent"
  },
  "overrides": []
},
"gridPos": {
  "h": 7,
  "w": 6,
  "x": 18,
  "y": 9
},
"id": 5,
"options": {
  "orientation": "auto",
  "reduceOptions": {
    "calcs": [
      "lastNotNull"
    ],
    "fields": "",

```

```

        "values": false
    },
    "showThresholdLabels": false,
    "showThresholdMarkers": true
},
"pluginVersion": "9.1.5",
"targets": [
    {
        "datasource": {
            "uid": "$datasource"
        },
        "expr": "100 - \\n(\\n  avg(node_memory_MemAvailable_bytes{job=\\\"prometheus-mysql-node-exporter\\\",
instance=\\\"$instance\\\"}) /\\n  avg(node_memory_MemTotal_bytes{job=\\\"prometheus-mysql-node-exporter\\\",
instance=\\\"$instance\\\"})\\n* 100\\n)\\n",
        "format": "time_series",
        "intervalFactor": 2,
        "legendFormat": "",
        "refId": "A"
    }
],
"title": "Memory Usage",
"type": "gauge"
},
{
    "collapsed": false,
    "datasource": {
        "type": "datasource",
        "uid": "grafana"
    },
    "gridPos": {
        "h": 1,
        "w": 24,
        "x": 0,
        "y": 16
    },
    "id": 12,
    "panels": [],
    "targets": [
        {
            "datasource": {
                "type": "datasource",
                "uid": "grafana"
            },
            "refId": "A"
        }
    ],
    "title": "Disk",
    "type": "row"
},
{
    "aliasColors": {},
    "bars": false,
    "dashLength": 10,
    "dashes": false,
    "datasource": {
        "uid": "$datasource"
    },
    "fill": 0,
    "fillGradient": 0,
    "gridPos": {
        "h": 7,
        "w": 12,
        "x": 0,
        "y": 17
    },
    "hiddenSeries": false,
    "id": 6,
    "legend": {
        "alignAsTable": false,
        "avg": false,
        "current": false,

```

```

    "max": false,
    "min": false,
    "rightSide": false,
    "show": true,
    "total": false,
    "values": false
  },
  "lines": true,
  "linewidth": 1,
  "links": [],
  "nullPointMode": "null",
  "options": {
    "alertThreshold": true
  },
  "percentage": false,
  "pluginVersion": "9.1.5",
  "pointradius": 5,
  "points": false,
  "renderer": "flot",
  "seriesOverrides": [
    {
      "alias": "/ read| written/",
      "yaxis": 1
    },
    {
      "alias": "/ io time/",
      "yaxis": 2
    }
  ],
  "spaceLength": 10,
  "stack": false,
  "steppedLine": false,
  "targets": [
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "rate(node_disk_read_bytes_total{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\", device=~\"(/dev/)?(mmcblk.p.+|nvme.+|rbd.+|sd.+|vd.+|xvd.+|dm- .+|dasd.+)\")}
[ $__rate_interval ])",
      "format": "time_series",
      "intervalFactor": 1,
      "legendFormat": "{{device}} read",
      "refId": "A"
    },
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "rate(node_disk_written_bytes_total{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\", device=~\"(/dev/)?(mmcblk.p.+|nvme.+|rbd.+|sd.+|vd.+|xvd.+|dm- .+|dasd.+)\")}
[ $__rate_interval ])",
      "format": "time_series",
      "intervalFactor": 1,
      "legendFormat": "{{device}} written",
      "refId": "B"
    },
    {
      "datasource": {
        "uid": "$datasource"
      },
      "expr": "rate(node_disk_io_time_seconds_total{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\", device=~\"(/dev/)?(mmcblk.p.+|nvme.+|rbd.+|sd.+|vd.+|xvd.+|dm- .+|dasd.+)\")}
[ $__rate_interval ])",
      "format": "time_series",
      "intervalFactor": 1,
      "legendFormat": "{{device}} io time",
      "refId": "C"
    }
  ],
  "thresholds": [],

```

```

"timeRegions": [],
"title": "Disk I/O",
"tooltip": {
  "shared": true,
  "sort": 0,
  "value_type": "individual"
},
"type": "graph",
"xaxis": {
  "mode": "time",
  "show": true,
  "values": []
},
"yaxes": [
  {
    "format": "Bps",
    "logBase": 1,
    "show": true
  },
  {
    "format": "percentunit",
    "logBase": 1,
    "show": true
  }
],
"yaxis": {
  "align": false
}
},
{
  "datasource": {
    "uid": "$datasource"
  },
  "fieldConfig": {
    "defaults": {
      "custom": {
        "align": "auto",
        "displayMode": "auto",
        "inspect": false
      },
      "mappings": [],
      "thresholds": {
        "mode": "absolute",
        "steps": [
          {
            "color": "green",
            "value": null
          },
          {
            "color": "yellow",
            "value": 0.8
          },
          {
            "color": "red",
            "value": 0.9
          }
        ]
      }
    },
    "unit": "decbytes"
  },
  "overrides": [
    {
      "matcher": {
        "id": "byName",
        "options": "Mounted on"
      },
      "properties": [
        {
          "id": "custom.width",
          "value": 260
        }
      ]
    }
  ]
}

```

```

    ]
  },
  {
    "matcher": {
      "id": "byName",
      "options": "Size"
    },
    "properties": [
      {
        "id": "custom.width",
        "value": 93
      }
    ]
  },
  {
    "matcher": {
      "id": "byName",
      "options": "Used"
    },
    "properties": [
      {
        "id": "custom.width",
        "value": 72
      }
    ]
  },
  {
    "matcher": {
      "id": "byName",
      "options": "Available"
    },
    "properties": [
      {
        "id": "custom.width",
        "value": 88
      }
    ]
  },
  {
    "matcher": {
      "id": "byName",
      "options": "Used, %"
    },
    "properties": [
      {
        "id": "unit",
        "value": "percentunit"
      },
      {
        "id": "custom.displayMode",
        "value": "gradient-gauge"
      },
      {
        "id": "max",
        "value": 1
      },
      {
        "id": "min",
        "value": 0
      }
    ]
  }
]
},
"gridPos": {
  "h": 7,
  "w": 12,
  "x": 12,
  "y": 17
},
"id": 7,

```

```

"options": {
  "footer": {
    "fields": "",
    "reducer": [
      "sum"
    ],
    "show": false
  },
  "showHeader": true
},
"pluginVersion": "9.1.5",
"targets": [
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "max by (mountpoint) (node_filesystem_size_bytes{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\", fstype!=\"\"})\n",
    "format": "table",
    "instant": true,
    "intervalFactor": 2,
    "legendFormat": "",
    "refId": "A"
  },
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "max by (mountpoint) (node_filesystem_avail_bytes{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\", fstype!=\"\"})\n",
    "format": "table",
    "instant": true,
    "intervalFactor": 2,
    "legendFormat": "",
    "refId": "B"
  }
],
"title": "Disk Space Usage",
"transformations": [
  {
    "id": "groupBy",
    "options": {
      "fields": {
        "Value #A": {
          "aggregations": [
            "lastNotNull"
          ],
          "operation": "aggregate"
        },
        "Value #B": {
          "aggregations": [
            "lastNotNull"
          ],
          "operation": "aggregate"
        },
        "mountpoint": {
          "aggregations": [],
          "operation": "groupby"
        }
      }
    }
  },
  {
    "id": "merge",
    "options": {}
  },
  {
    "id": "calculateField",
    "options": {
      "alias": "Used",
      "binary": {

```

```

        "left": "Value #A (lastNotNull)",
        "operator": "-",
        "reducer": "sum",
        "right": "Value #B (lastNotNull)"
    },
    "mode": "binary",
    "reduce": {
        "reducer": "sum"
    }
},
{
    "id": "calculateField",
    "options": {
        "alias": "Used, %",
        "binary": {
            "left": "Used",
            "operator": "/",
            "reducer": "sum",
            "right": "Value #A (lastNotNull)"
        },
        "mode": "binary",
        "reduce": {
            "reducer": "sum"
        }
    }
},
{
    "id": "organize",
    "options": {
        "excludeByName": {},
        "indexByName": {},
        "renameByName": {
            "Value #A (lastNotNull)": "Size",
            "Value #B (lastNotNull)": "Available",
            "mountpoint": "Mounted on"
        }
    }
},
{
    "id": "sortBy",
    "options": {
        "fields": {},
        "sort": [
            {
                "field": "Mounted on"
            }
        ]
    }
},
],
"type": "table"
},
{
    "collapsed": false,
    "datasource": {
        "type": "datasource",
        "uid": "grafana"
    },
    "gridPos": {
        "h": 1,
        "w": 24,
        "x": 0,
        "y": 24
    },
    "id": 13,
    "panels": [],
    "targets": [
        {
            "datasource": {
                "type": "datasource",

```

```

        "uid": "grafana"
    },
    "refId": "A"
}
],
"title": "Network",
"type": "row"
},
{
    "aliasColors": {},
    "bars": false,
    "dashLength": 10,
    "dashes": false,
    "datasource": {
        "uid": "$datasource"
    },
    "description": "Network received (bits/s)",
    "fill": 0,
    "fillGradient": 0,
    "gridPos": {
        "h": 7,
        "w": 12,
        "x": 0,
        "y": 25
    },
    "hiddenSeries": false,
    "id": 8,
    "legend": {
        "alignAsTable": false,
        "avg": false,
        "current": false,
        "max": false,
        "min": false,
        "rightSide": false,
        "show": true,
        "total": false,
        "values": false
    },
    "lines": true,
    "linewidth": 1,
    "links": [],
    "nullPointMode": "null",
    "options": {
        "alertThreshold": true
    },
    "percentage": false,
    "pluginVersion": "9.1.5",
    "pointradius": 5,
    "points": false,
    "renderer": "flot",
    "seriesOverrides": [],
    "spaceLength": 10,
    "stack": false,
    "steppedLine": false,
    "targets": [
        {
            "datasource": {
                "uid": "$datasource"
            },
            "expr": "rate(node_network_receive_bytes_total{job=\"prometheus-mysql-node-exporter\", instance=\"$instance\", device!=\"lo\"}[$__rate_interval]) * 8",
            "format": "time_series",
            "intervalFactor": 1,
            "legendFormat": "{{device}}",
            "refId": "A"
        }
    ],
    "thresholds": [],
    "timeRegions": [],
    "title": "Network Received",
    "tooltip": {

```

```

    "shared": true,
    "sort": 0,
    "value_type": "individual"
  },
  "type": "graph",
  "xaxis": {
    "mode": "time",
    "show": true,
    "values": []
  },
  "yaxes": [
    {
      "format": "bps",
      "logBase": 1,
      "min": 0,
      "show": true
    },
    {
      "format": "bps",
      "logBase": 1,
      "min": 0,
      "show": true
    }
  ],
  "yaxis": {
    "align": false
  }
},
{
  "aliasColors": {},
  "bars": false,
  "dashLength": 10,
  "dashes": false,
  "datasource": {
    "uid": "$datasource"
  },
  "description": "Network transmitted (bits/s)",
  "fill": 0,
  "fillGradient": 0,
  "gridPos": {
    "h": 7,
    "w": 12,
    "x": 12,
    "y": 25
  },
  "hiddenSeries": false,
  "id": 9,
  "legend": {
    "alignAsTable": false,
    "avg": false,
    "current": false,
    "max": false,
    "min": false,
    "rightSide": false,
    "show": true,
    "total": false,
    "values": false
  },
  "lines": true,
  "linewidth": 1,
  "links": [],
  "nullPointMode": "null",
  "options": {
    "alertThreshold": true
  },
  "percentage": false,
  "pluginVersion": "9.1.5",
  "pointradius": 5,
  "points": false,
  "renderer": "flot",
  "seriesOverrides": [],

```

```

"spaceLength": 10,
"stack": false,
"steppedLine": false,
"targets": [
  {
    "datasource": {
      "uid": "$datasource"
    },
    "expr": "rate(node_network_transmit_bytes_total{job=\"prometheus-mysql-node-exporter\",
instance=\"$instance\", device!=\"lo\"}[$__rate_interval]) * 8",
    "format": "time_series",
    "intervalFactor": 1,
    "legendFormat": "{{device}}",
    "refId": "A"
  }
],
"thresholds": [],
"timeRegions": [],
"title": "Network Transmitted",
"tooltip": {
  "shared": true,
  "sort": 0,
  "value_type": "individual"
},
"type": "graph",
"xaxis": {
  "mode": "time",
  "show": true,
  "values": []
},
"yaxes": [
  {
    "format": "bps",
    "logBase": 1,
    "min": 0,
    "show": true
  },
  {
    "format": "bps",
    "logBase": 1,
    "min": 0,
    "show": true
  }
],
"yaxis": {
  "align": false
}
},
"refresh": "30s",
"schemaVersion": 37,
"style": "dark",
"tags": [
  "prometheus-mysql-node-exporter-mixin"
],
"templating": {
  "list": [
    {
      "current": {
        "selected": false,
        "text": "Prometheus",
        "value": "Prometheus"
      },
      "hide": 0,
      "includeAll": false,
      "label": "Data Source",
      "multi": false,
      "name": "datasource",
      "options": [],
      "query": "prometheus",
      "refresh": 1,

```

```

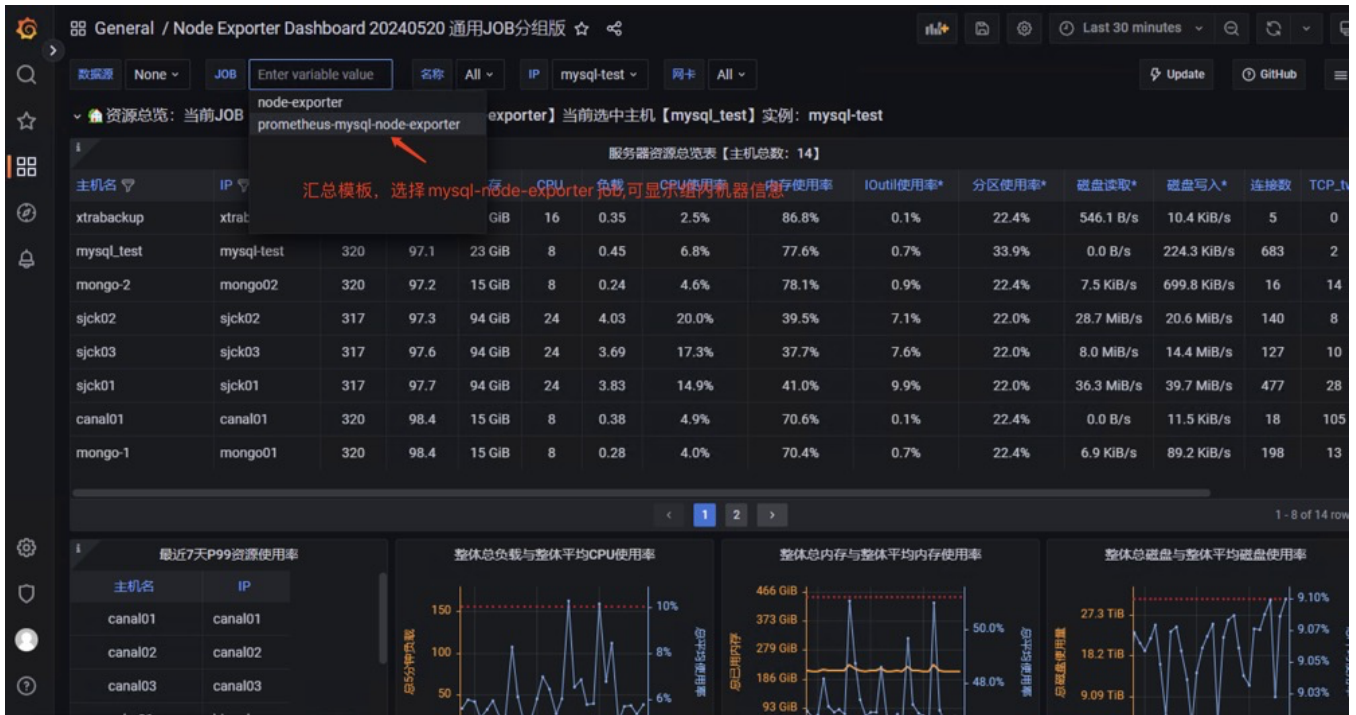
        "regex": "",
        "skipUrlSync": false,
        "type": "datasource"
    },
    {
        "current": {
            "selected": false,
            "text": "10.20.41.50:9796",
            "value": "10.20.41.50:9796"
        },
        "datasource": {
            "type": "prometheus",
            "uid": "$datasource"
        },
        "definition": "",
        "hide": 0,
        "includeAll": false,
        "label": "Instance",
        "multi": false,
        "name": "instance",
        "options": [],
        "query": {
            "query": "label_values(node_uname_info{job=\"prometheus-mysql-node-exporter\", sysname!=\"Darwin\"}, instance)",
            "refId": "Prometheus-instance-Variable-Query"
        },
        "refresh": 2,
        "regex": "",
        "skipUrlSync": false,
        "sort": 0,
        "tagValuesQuery": "",
        "tagsQuery": "",
        "type": "query",
        "useTags": false
    }
]
},
"time": {
    "from": "now-1h",
    "to": "now"
},
"timepicker": {
    "refresh_intervals": [
        "5s",
        "10s",
        "30s",
        "1m",
        "5m",
        "15m",
        "30m",
        "1h",
        "2h",
        "1d"
    ],
    "time_options": [
        "5m",
        "15m",
        "1h",
        "6h",
        "12h",
        "24h",
        "2d",
        "7d",
        "30d"
    ]
},
"timezone": "browser",
"title": "MySQL-Nodes-Exporter",
"uid": "5KlmGKsql",
"version": 1,
"weekStart": ""

```

}



专用模板 不过此模板磁盘空间是字节/1000计算的 导致显示空间变大, 小问题



汇总模板, 选择mysql-node-exporter job, 可显示组内机器信息